LR(0) Parser — FIFA Player Card Language

Theory of Automata • Context-Free Grammar & Parsing Table

Language Specification

The language models a **FIFA-style player card**. Each record contains a player name, overall rating, position, country, club, and a sequence of attribute stats.

Token Classes (Terminals)

Terminal	Regex	Examples
NAME	^[A-Za-z][A-Za-z .'-]*$	Lionel Messi, M. Salah
RATING	^[1-9][0-9]?$	93, 7
POS	^(ST|CF|CAM|CM|CDM|LW|RW|...)$	ST, CAM, GK
COUNTRY	^[A-Za-z][A-Za-z .-]*$	Argentina, Brazil
CLUB	^[A-Za-z0-9][A-Za-z0-9 .&'-]*$	Inter Miami, Man City
LABEL	^(PAC|SHO|PAS|DRI|DEF|PHY)$	PAC, DEF
VAL	^[1-9][0-9]?$	85, 72

Context-Free Grammar

Augmented Grammar

(0)	$S' \rightarrow S$	Augmented start rule
(1)	$S \rightarrow$ *player_entry*	Top-level rule
(2)	*player_entry* $\rightarrow$ NAME RATING POS COUNTRY CLUB *stat_list*	Full player record
(3)	*stat_list* $\rightarrow$ *stat_list stat*	Multiple stats (left-recursive)
(4)	*stat_list* $\rightarrow$ *stat*	Single stat (base case)
(5)	*stat* $\rightarrow$ LABEL VAL	e.g. PAC 85

LR(0) Canonical Collection

Each state contains a set of *LR(0) items* (productions with a • marking the parser's current position). The closure adds all productions for non-terminals immediately following a •.

State I_0 — Initial state

$S' \to \bullet\, S$
$S \to \bullet\, \mathit{player_entry}$
$\mathit{player_entry} \to \bullet\, \texttt{NAME RATING POS COUNTRY CLUB}\ \mathit{stat_list}$

State I_1 — After S

$S' \to S\, \bullet$

State I_2 — After *player_entry*

$S \to \mathit{player_entry}\, \bullet$

State I_3 — After NAME

$\mathit{player_entry} \to \texttt{NAME} \bullet \texttt{RATING POS COUNTRY CLUB}\ \mathit{stat_list}$

State I_4 — After RATING

$\mathit{player_entry} \to \texttt{NAME RATING} \bullet \texttt{POS COUNTRY CLUB}\ \mathit{stat_list}$

State I_5 — After POS

$\mathit{player_entry} \to \texttt{NAME RATING POS} \bullet \texttt{COUNTRY CLUB}\ \mathit{stat_list}$

State I_6 — After COUNTRY

$\mathit{player_entry} \to \texttt{NAME} \ldots \texttt{COUNTRY} \bullet \texttt{CLUB}\ \mathit{stat_list}$

State I_7 — After CLUB (closure)

$\mathit{player_entry} \to \ldots \texttt{CLUB} \bullet \mathit{stat_list}$
$\mathit{stat_list} \to \bullet\, \mathit{stat_list}\ \mathit{stat}$
$\mathit{stat_list} \to \bullet\, \mathit{stat}$
$\mathit{stat} \to \bullet\, \texttt{LABEL VAL}$

State I_8 — After *stat_list* (closure)

$\mathit{player_entry} \to \ldots \texttt{CLUB}\ \mathit{stat_list}\, \bullet$
$\mathit{stat_list} \to \mathit{stat_list} \bullet \mathit{stat}$
$\mathit{stat} \to \bullet\, \texttt{LABEL VAL}$

State I_9 — After single *stat*

$\mathit{stat_list} \to \mathit{stat}\, \bullet$

State I_{10} — After LABEL

$\mathit{stat} \to \texttt{LABEL} \bullet \texttt{VAL}$

State I_{11} — After LABEL VAL

$\mathit{stat} \to \texttt{LABEL VAL}\, \bullet$

State I_{12} — After *stat_list stat*

$\mathit{stat_list} \to \mathit{stat_list}\ \mathit{stat}\, \bullet$

LR(0) Automaton (DFA Transitions)

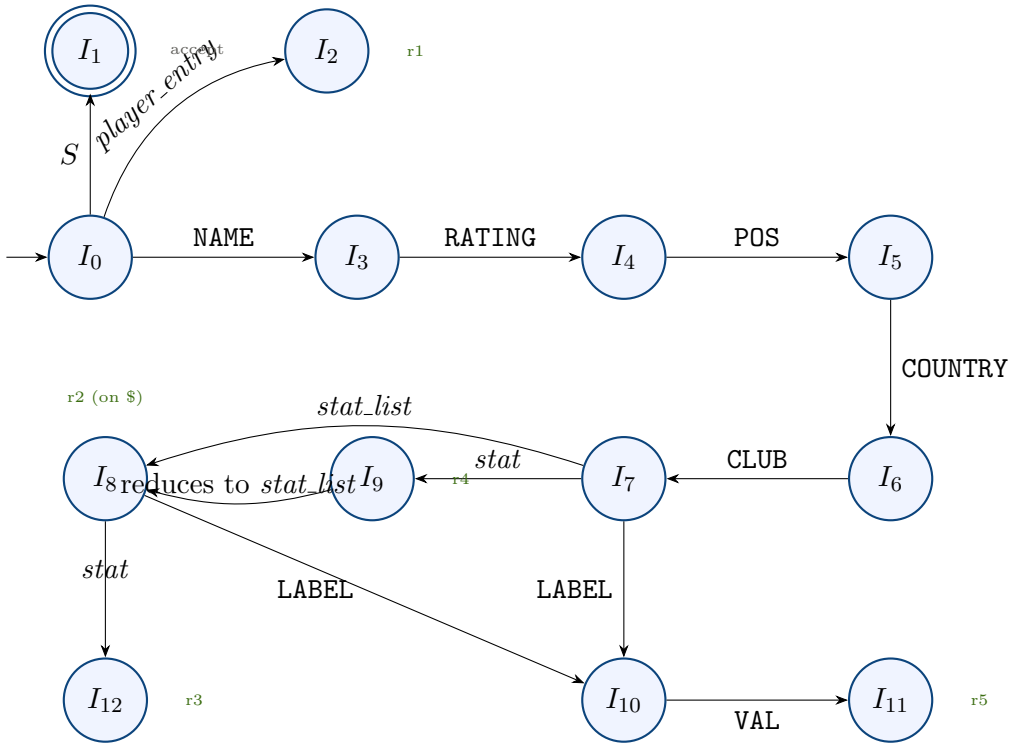

Figure 1: LR(0) DFA. Double-bordered state I_1 is the accept state. States annotated with **rN** perform a reduction in the parsing table.

LR(0) Parsing Table

State	ACTION								GOTO			
	NAME	RATING	POS	COUNTRY	CLUB	LABEL	VAL	$	S	*p_entry*	*s_list*	*stat*
0	d3								1	2		
1								acc				
2	r1	r1	r1	r1	r1	r1	r1	r1				
3		d4										
4			d5									
5				d6								
6					d7							
7						d10					8	9
8	r2	r2	r2	r2	r2	d10	r2	r2				12
9	r4	r4	r4	r4	r4	r4	r4	r4				
10							d11					
11	r5	r5	r5	r5	r5	r5	r5	r5				
12	r3	r3	r3	r3	r3	r3	r3	r3				

Legend: dN = shift and go to state N rN = reduce by rule N acc = accept N (GOTO) = go to state N

Note: Column headers abbreviated — *p_entry* = *player_entry*, *s_list* = *stat_list*. No shift/reduce or reduce/reduce conflicts exist: this grammar **is** LR(0).

Parsing Trace Example

Input string: `NAME RATING POS COUNTRY CLUB LABEL VAL $`

Stack	Remaining Input	Action
0	NAME RATING POS COUNTRY CLUB LABEL VAL $	**shift** $\rightarrow I_3$
0 NAME 3	RATING POS COUNTRY CLUB LABEL VAL $	**shift** $\rightarrow I_4$
0 NAME 3 RATING 4	POS COUNTRY CLUB LABEL VAL $	**shift** $\rightarrow I_5$
0 ... POS 5	COUNTRY CLUB LABEL VAL $	**shift** $\rightarrow I_6$
0 ... COUNTRY 6	CLUB LABEL VAL $	**shift** $\rightarrow I_7$
0 ... CLUB 7	LABEL VAL $	**shift** $\rightarrow I_{10}$
0 ... LABEL 10	VAL $	**shift** $\rightarrow I_{11}$
0 ... LABEL 10 VAL 11	$	**r5**: *stat* $\rightarrow$ LABEL VAL
0 ... CLUB 7 stat 9	$	**r4**: *stat_list* $\rightarrow$ *stat*
0 ... CLUB 7 stat_list 8	$	**r2**: *player_entry* $\rightarrow$ NAME ... *stat_list*
0 player_entry 2	$	**r1**: $S \rightarrow$ *player_entry*
0 S 1	$	**accept** ✓

For multiple stats (e.g. `LABEL VAL LABEL VAL`): after the first r5+r4 cycle the parser is in I_8, where reading another `LABEL` loops through $I_{10} \rightarrow I_{11}$, reducing by r5 then r3 for each additional stat.